



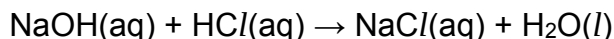
## 2017 MJC H2 Chemistry Prelim Paper 4

- 1** Magnesium carbonate is a white solid that is readily found in nature. In this experiment, you will determine the percentage purity of a sample impure magnesium carbonate.

**FA 1** is a sample of impure magnesium carbonate. You may assume that only the magnesium carbonate present in the sample will react with the acid. You will first dissolve **FA 1** in excess hydrochloric acid.



The resulting solution will then be titrated using sodium hydroxide. You may assume that no compounds present in **FA 1** will react with the sodium hydroxide.



**FA 1** is impure magnesium carbonate.

**FA 2** is  $2.0 \text{ mol dm}^{-3}$  hydrochloric acid,  $\text{HCl}$ .

**FA 4** is  $0.20 \text{ mol dm}^{-3}$  sodium hydroxide,  $\text{NaOH}$ .  
methyl orange indicator

### (a) Procedure

#### Preparation of standard solution

1. Fill a burette with **FA 2**.
  2. Run between  $47.50$  and  $48.50 \text{ cm}^3$  of **FA 2** into a  $250 \text{ cm}^3$  beaker.
  3. Record your burette readings and the volume of **FA 2** in the space below.
- 
4. Weigh accurately about  $2.0 \text{ g}$  of **FA 1** in a weighing bottle.
  5. Transfer the **FA 1** from the weighing bottle as fully as you can into the beaker containing **FA 2**.
  6. Reweigh the weighing bottle and record all masses in the space below.

7. Stir the mixture carefully until all the solid has reacted.
8. Transfer the contents of the beaker into the volumetric flask.
9. Rinse the beaker with distilled water and add it to the volumetric flask. Make the solution up to 250 cm<sup>3</sup> with distilled water and mix thoroughly. Label this solution as **FA 3**.

#### Titration

1. Fill another burette with **FA 3** from the volumetric flask.
2. Pipette 25.0 cm<sup>3</sup> of **FA 4** into a conical flask.
3. Add a few drops of methyl orange indicator.
4. Titrate **FA 4** with **FA 3** until the end-point is reached.
5. Repeat the titration as many times as you think necessary to obtain consistent results.
6. Record your titration results in the space provided below.

[7]

| M1 | M2 | M3 | M4 | M5 | M6 | M7 |
|----|----|----|----|----|----|----|
|    |    |    |    |    |    |    |

- (b) From your titration results, obtain a suitable volume of **FA 3** to be used in your calculations. Show clearly how you obtained this volume.

volume of **FA 3** = ..... [1]

**(c) Calculations**

- (i)** Calculate the amount of hydrochloric acid present in the volume of **FA 3** obtained in **(b)**.

amount of  $\text{HCl}$  = .....[1]

- (ii)** Calculate the amount of hydrochloric acid present in  $250\text{ cm}^3$  of **FA 3**.

amount of  $\text{HCl}$  in  $250\text{ cm}^3$  **FA 3** = .....[1]

- (iii)** Calculate the amount of hydrochloric acid in **FA 2** that has reacted with magnesium carbonate in **FA 1**.

amount of  $\text{HCl}$  reacted with  $\text{MgCO}_3$  = .....[1]

- (iv) Calculate the percentage purity of magnesium carbonate present in **FA 1**.  
[Ar: C, 12.0; O, 16.0; Mg, 24.3]

percentage purity of **FA1** = ..... [1]

- (d) (i) The maximum error in a single burette reading is  $\pm 0.05 \text{ cm}^3$ .

Student X, carrying out this experiment, recorded that  $48.50 \text{ cm}^3$  of **FA 2** was added to **FA 1**.

What are the smallest and largest possible volumes of **FA 2** that were added?

smallest volume used = .....

largest volume used = .....

[1]

- (ii) Student Y used an identical mass of **FA 1** but added  $47.70 \text{ cm}^3$  of **FA 2** instead. How would the percentage purity of **FA 1** calculated by student Y compare to that obtained by student X? Explain your answer.

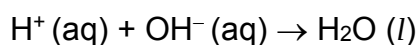
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[1]

[Total: 14]

## 2 Determination of the enthalpy change of neutralisation, $\Delta H_n$

The enthalpy change of neutralisation,  $\Delta H_n$ , is the enthalpy change when one mole of water is formed during a neutralisation reaction as shown in the equation.



**FA 5** is a solution of sulfuric acid,  $\text{H}_2\text{SO}_4$

**FA 6** is  $1.50 \text{ mol dm}^{-3}$  potassium hydroxide,  $\text{KOH}$

You will perform a series of experiments using different volumes of **FA 5** and **FA 6**. The change in temperature,  $\Delta T$ , for each experiment will be determined and used to plot a graph of  $\Delta T$  against volume of **FA 5** used.

You will then use data from the graph to determine a value for the enthalpy change of neutralisation,  $\Delta H_n$ .

### (a) Determining the change in temperature for a series of reactions between **FA 5** and **FA 6**

#### (i) Procedure

1. Support the Styrofoam cup in a  $250 \text{ cm}^3$  beaker.
2. Add  $10.0 \text{ cm}^3$  of **FA 5** from a measuring cylinder into the Styrofoam cup.
3. Measure the temperature of **FA 5** in the Styrofoam cup. Record the initial temperature of the solution of **FA 5** as  $T_{\text{FA5}}$ .
4. Wash and dry the thermometer.
5. Measure  $40.0 \text{ cm}^3$  of **FA 6** into another measuring cylinder.
6. Measure the temperature of **FA 6** solution using the thermometer and record the initial temperature of the solution of **FA 6** as  $T_{\text{FA6}}$ .
7. Tip the **FA 6** in the measuring cylinder into the Styrofoam cup, stir and record the maximum temperature obtained in the reaction as  $T_{\text{max}}$ .
8. Rinse and dry the Styrofoam cup and the thermometer.
9. Repeat steps 2 to 8 using  $20.0 \text{ cm}^3$  and  $30.0 \text{ cm}^3$  of **FA 5** and appropriate volumes of **FA 6** each time such that the total volume of the reacting mixture is  $50 \text{ cm}^3$ .

In an appropriate format in the space provided, record:

- all measurements of volumes used,
- all temperatures measured and the change in temperature,  $\Delta T$ .

$$\Delta T = T_{\text{max}} - T_{\text{weighted initial}}$$

$$T_{\text{weighted initial}} = \frac{(\text{Volume of FA 5} \times T_{\text{FA 5}}) + (\text{Volume of FA 6} \times T_{\text{FA 6}})}{\text{Volume of FA 5} + \text{Volume of FA 6}}$$

These data should be recorded for the experiments described in **(a)(i)**, together with additional experiments described in **(a)(ii)**.

## Results

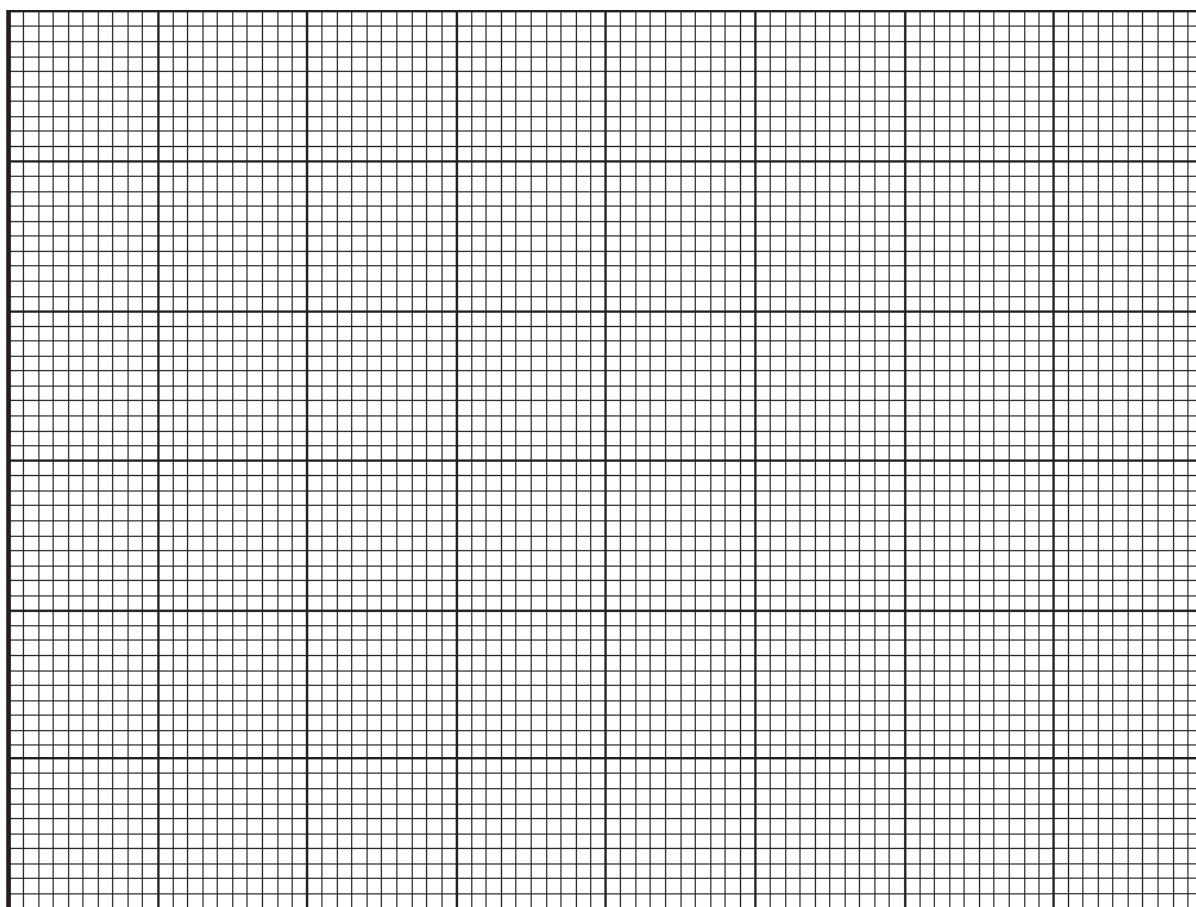
[5]

| M1 | M2 | M3 | M4 | M5 |
|----|----|----|----|----|
|    |    |    |    |    |

- (a)(ii)** On the grid provided, plot a graph of  $\Delta T$  (y-axis) against volume of **FA 5** (x-axis) using the data you obtained in **(a)(i)**.

Consider your plotted plots. Carry out four more experiments which will enable you to identify the volume of **FA 5** which gives the maximum change in temperature,  $\Delta T_{max}$ . In each case, the total volume of the reaction mixture must be 50 cm<sup>3</sup>.

You may find it useful to consider the results obtained from each experiment before choosing the volumes of solutions for the next experiment.



- (iii)** Draw a line of best fit through the points where the temperature rise is increasing and another best fit line through the points where the temperature rise is decreasing. Extrapolate these two lines until they cross.

[3]

| M1 | M2 | M3 |
|----|----|----|
|    |    |    |

- (iv)** Determine from your graph, the maximum change in temperature,  $\Delta T_{max}$ , and the volume,  $V_{max}$ , of **FA 5** required to obtain this value.

$\Delta T_{max} = \dots\dots\dots$

$V_{max} = \dots\dots\dots$

[2]

**(b)** Using your answers in **(a)(iv)**, calculate

**(i)** the concentration, in  $\text{mol dm}^{-3}$ , of  $\text{H}_2\text{SO}_4$  in **FA 5**.

concentration of  $\text{H}_2\text{SO}_4$  in **FA 5** = ..... [1]

**(ii)** the heat change for the neutralisation reaction at  $\Delta T_{\text{max}}$ .

You should assume that the specific heat capacity of the final solution is  $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ , and the density of the final solution is  $1.00 \text{ g cm}^{-3}$ .

heat change = ..... [1]

**(c)** Using your answers from **(b)(i)** and **(b)(ii)**, calculate a value for the enthalpy change of neutralisation,  $\Delta H_n$ .

$\Delta H_n = \dots\dots\dots$  [3]

| M1 | M2 | M3 |
|----|----|----|
|    |    |    |



- (d) When the experiment is done in the way described, the results are not very accurate.

Apart from limitations due to the accuracy of the measuring equipment, suggest why the temperature rises are more inaccurate as they approach their maximum value.

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[1]

- (e) In another experiment,  $1.50 \text{ mol dm}^{-3}$  aqueous ammonia is used instead of the  $1.50 \text{ mol dm}^{-3}$  aqueous potassium hydroxide.

Draw on your graph another pair of lines to show the results you would expect to obtain. Explain your answer.

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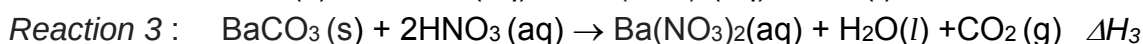
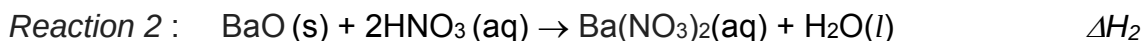
[2]

**(f) Planning**

Barium carbonate,  $\text{BaCO}_3$ , is widely used in the ceramics industry as an ingredient in glazes.  $\text{BaCO}_3$  can be formed from the reaction between barium oxide and carbon dioxide as shown in *Reaction 1*.



Experimentally, it is difficult to measure the enthalpy change of reaction,  $\Delta H_1$  directly. Instead, Hess's Law can be used to calculate  $\Delta H_1$  using experimentally determined values of  $\Delta H_2$  and  $\Delta H_3$  for the following reactions as shown below.



- (i)** Construct an energy cycle using the information in the question to show how the enthalpy change of reaction,  $\Delta H_1$ , can be calculated from  $\Delta H_2$  and  $\Delta H_3$ .

[1]

- (ii) You are to plan an experiment that will allow you to determine the enthalpy change of *Reaction 2*,  $\Delta H_2$ . Your plan must enable you to plot a graph that includes data from the experiment. You may assume that all solutions have a density of  $1 \text{ g cm}^{-3}$  and the specific heat capacity of water is  $4.18 \text{ J g}^{-1} \text{ K}^{-1}$ .

You may also assume that the following are provided:

- 3 g solid barium oxide
- $120 \text{ cm}^3$  of  $0.500 \text{ mol dm}^{-3}$  aqueous nitric acid,  $\text{HNO}_3$
- Styrofoam cup
- stop-watch
- thermometer
- apparatus and materials normally found in a school laboratory.

**YOU ARE NOT REQUIRED TO PERFORM THIS EXPERIMENT.**

In your plan, you should include details of:

- calculations of suitable quantities of reagents used;
- an outline of all essential experimental procedures;
- a sketch of the graph you expect to obtain and
- an outline of the calculations you would perform to find  $\Delta H_2$  in  $\text{kJ mol}^{-1}$ .

[6]

[Total: 25]

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### 3 Inorganic and Organic Analysis

(a) You are provided with **FA 7** and **FA 8**.

In all tests, the reagents should be added gradually until no further change is observed, with shaking after each addition.

At each stage of any test you are to record details of the following.

- colour changes seen
- the formation of any precipitate
- the solubility of such precipitates in an excess of the reagent added

Where gases are released they should be identified by a test, described in the appropriate place in your observations.

Record your observations in the spaces provided. You should indicate clearly at which stage in a test a change occurs.

1. **FA 7** is a sodium compound containing one anion listed on page 20. Dissolve the **FA 7** provided in about 15 cm<sup>3</sup> of distilled water in a boiling tube.

Carry out the following tests and record your observations in the table below.

| Test                                                                                                                                                    | Observations |
|---------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| (i) To a 1 cm depth of the solution of <b>FA 7</b> in a test-tube, add a few drops of aqueous barium nitrate followed by dilute nitric acid.            |              |
| (ii) To a 1 cm depth of the solution of <b>FA 7</b> in a test-tube, add an equal volume of aqueous hydrogen peroxide, then                              |              |
| add a few drops of aqueous barium nitrate, followed by dilute nitric acid.                                                                              |              |
| (iii) To a 2 cm depth of the solution of <b>FA 7</b> in a test tube, add an equal volume of sulfuric acid, then heat the mixture gently and cautiously. |              |

| Test                                                                                                                      | Observations |
|---------------------------------------------------------------------------------------------------------------------------|--------------|
| (iv) To a 1 cm depth of the solution of <b>FA 7</b> in a test-tube, add an equal volume of aqueous sodium hydroxide, then |              |
| add a few drops of aqueous potassium manganate(VII),                                                                      |              |
| followed by sulfuric acid.                                                                                                |              |

(v) Identify the anion in **FA 7**, and state the evidences for your identification.

anion .....

evidence .....

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[6]

2. **FA 8** solution contains two cations from those listed in the Qualitative Analysis Notes on page 19. Carry out tests using reagents that will allow you to identify which ions are present.

- (i) In the space below, draw a single table to record details of the tests performed and observations made.

[4]

- (ii) Identify the cations in **FA 8**, and state the evidences for your identification.

cation .....

evidence .....

.....

.....

cation .....

evidence .....

.....

.....

[2]

(b) You are provided with an organic solution **FA 9** which contains one functional group.

**FA 9** gives a positive test with 2,4-dinitrophenylhydrazine. Devise two other confirmatory tests using the **bench reagents provided** to identify the functional group present in **FA 9**.

(i) In the space below, draw a single table to record details of the tests performed and observations made.

(ii) Identify the functional group in **FA 9**, and state the evidences for your identification.

Functional group in **FA 9** .....

evidence .....

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.....

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[4]

[Total : 16]



